



Triangle wave

A **triangular wave** or **triangle wave** is a non-sinusoidal waveform named for its triangular shape. It is a periodic, piecewise linear, continuous real function.

Like a square wave, the triangle wave contains only odd harmonics. However, the higher harmonics roll off much faster than in a square wave (proportional to the inverse square of the harmonic number as opposed to just the inverse).

Definitions

Definition

A triangle wave of period p that spans the range $[0, 1]$ is defined as

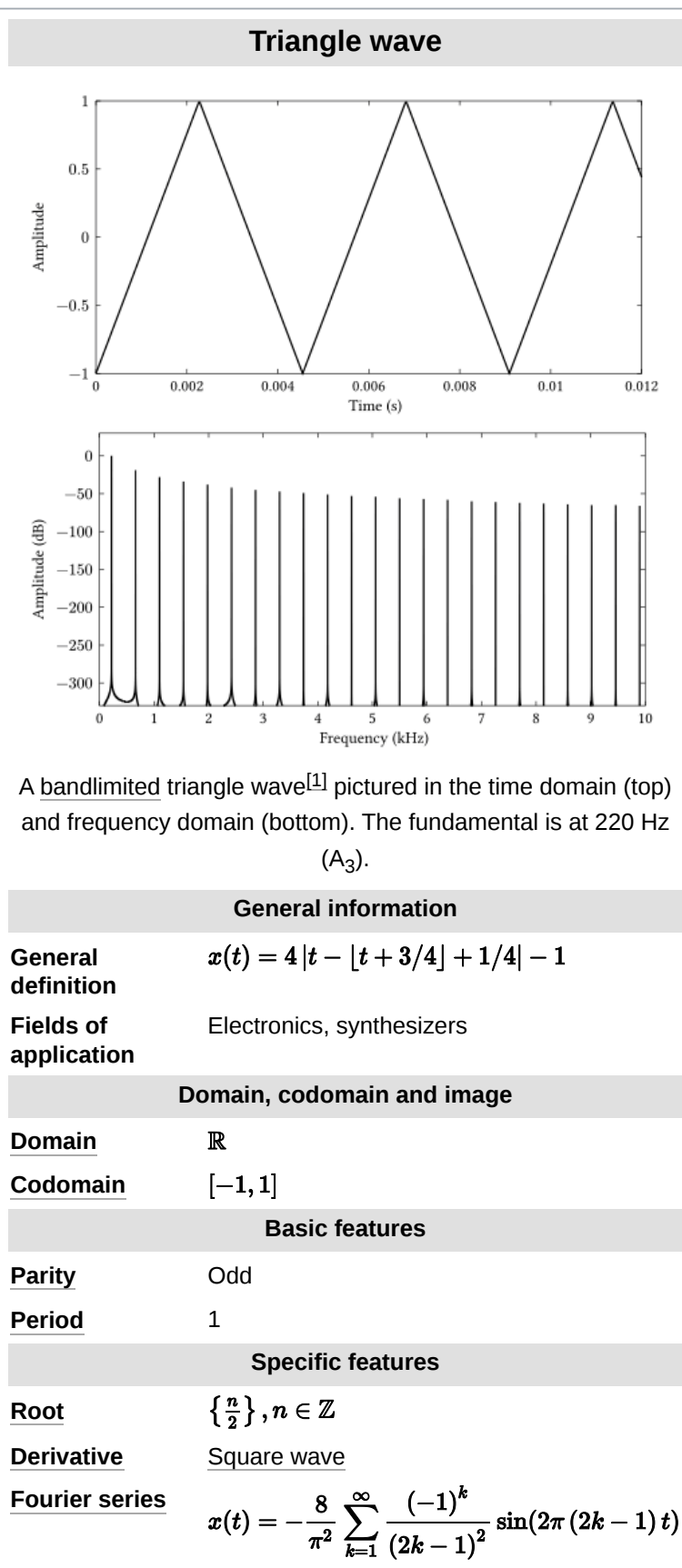
$$x(t) = 2 \left| \frac{t}{p} - \left\lfloor \frac{t}{p} + \frac{1}{2} \right\rfloor \right|,$$

where $\lfloor \rfloor$ is the floor function. This can be seen to be the absolute value of a shifted sawtooth wave.

For a triangle wave spanning the range $[-1, 1]$ the expression becomes

$$x(t) = 2 \left| 2 \left(\frac{t}{p} - \left\lfloor \frac{t}{p} + \frac{1}{2} \right\rfloor \right) \right| - 1.$$

A more general equation for a triangle wave with amplitude a and period p using the modulo operation and absolute value is



$$y(x) = \frac{4a}{p} \left| \left(\left(x - \frac{p}{4} \right) \bmod p \right) - \frac{p}{2} \right| - a.$$

For example, for a triangle wave with amplitude 5 and period 4:

$$y(x) = 5 \left| \left((x - 1) \bmod 4 \right) - 2 \right| - 5.$$

A phase shift can be obtained by altering the value of the $-p/4$ term, and the vertical offset can be adjusted by altering the value of the $-a$ term.

As this only uses the modulo operation and absolute value, it can be used to simply implement a triangle wave on hardware electronics.

Note that in many programming languages, the % operator is a remainder operator (with result the same sign as the dividend), not a modulo operator; the modulo operation can be obtained by using $((x \% p) + p) \% p$ in place of $x \% p$. In e.g. JavaScript, this results in an equation of the form $4*a/p * \text{Math.abs}(((x - p/4) \% p) + p) \% p - p/2) - a$.

Relation to the square wave

The triangle wave can also be expressed as the integral of the square wave:

$$x(t) = \int_0^t \text{sgn} \left(\sin \frac{u}{p} \right) du.$$

Expression in trigonometric functions

A triangle wave with period p and amplitude a can be expressed in terms of sine and arcsine (whose value ranges from $-\pi/2$ to $\pi/2$):

$$y(x) = \frac{2a}{\pi} \arcsin \left(\sin \left(\frac{2\pi}{p} x \right) \right).$$

The identity $\cos x = \sin \left(\frac{p}{4} - x \right)$ can be used to convert from a triangle "sine" wave to a triangular "cosine" wave. This phase-shifted triangle wave can also be expressed with cosine and arccosine:

$$y(x) = a - \frac{2a}{\pi} \arccos \left(\cos \left(\frac{2\pi}{p} x \right) \right).$$

Expressed as alternating linear functions

Another definition of the triangle wave, with range from -1 to 1 and period p , is

$$x(t) = \frac{4}{p} \left(t - \frac{p}{2} \left\lfloor \frac{2t}{p} + \frac{1}{2} \right\rfloor \right) (-1)^{\left\lfloor \frac{2t}{p} + \frac{1}{2} \right\rfloor}.$$

Triangle wave sound sample



0:00 / 0:00

5 seconds of triangle wave at 220 Hz

[Problems playing this file? See media help.](#)

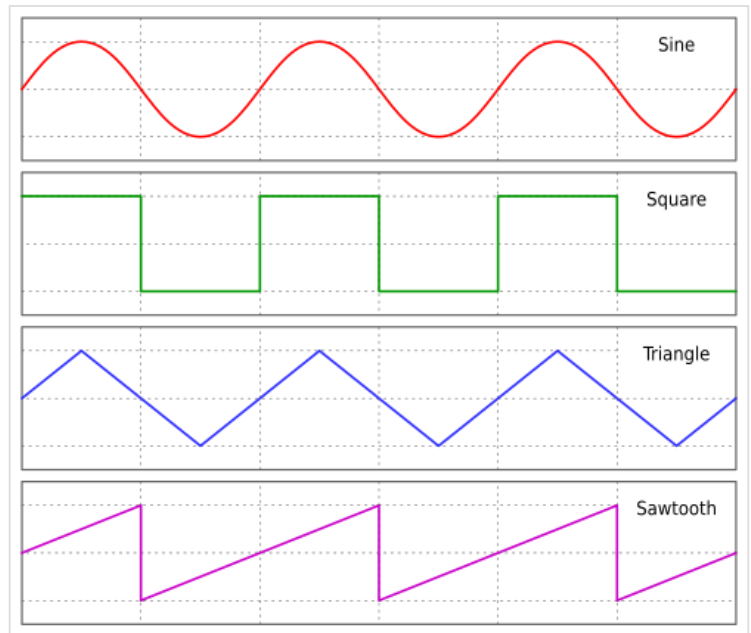
Additive Triangle wave sound sample



0:00 / 0:00

After each second, a harmonic is added to a sine wave creating a triangle 220 Hz wave

[Problems playing this file? See media help.](#)



Sine, square, triangle, and sawtooth waveforms

Harmonics

It is possible to approximate a triangle wave with additive synthesis by summing odd harmonics of the fundamental while multiplying every other odd harmonic by -1 (or, equivalently, changing its phase by π) and multiplying the amplitude of the harmonics by one over the square of their mode number, n (which is equivalent to one over the square of their relative frequency to the fundamental).

The above can be summarised mathematically as follows:

$$x_{\text{triangle}}(t) = \frac{8}{\pi^2} \sum_{i=0}^{N-1} \frac{(-1)^i}{n^2} \sin(2\pi f_0 n t),$$

where N is the number of harmonics to include in the approximation, t is the independent variable (e.g. time for sound waves), f_0 is the fundamental frequency, and i is the harmonic label which is related to its mode number by $n = 2i + 1$.

This infinite Fourier series converges quickly to the triangle wave as N tends to infinity, as shown in the animation.

Arc length

The arc length per period for a triangle wave, denoted by s , is given in terms of the amplitude a and period length p by

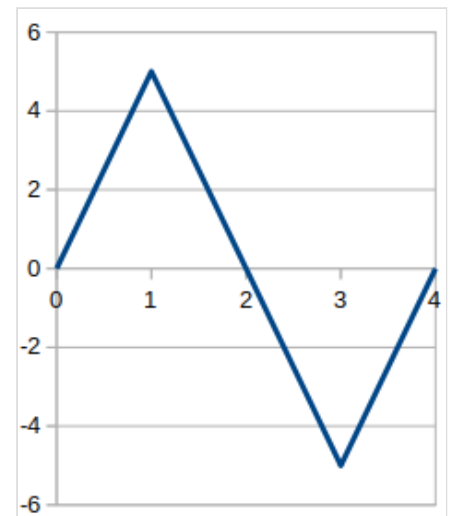
$$s = \sqrt{(4a)^2 + p^2}.$$

See also

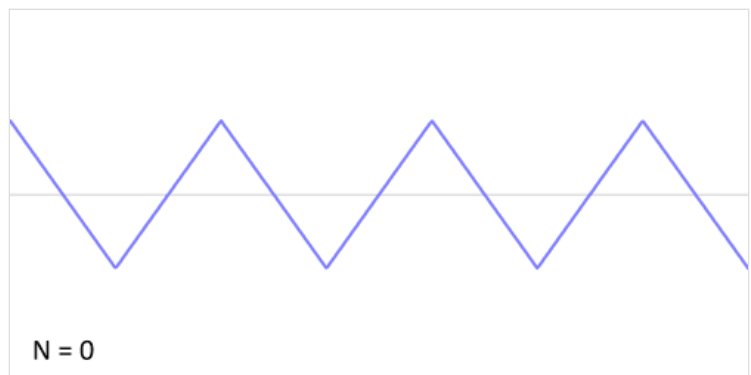
- List of periodic functions
- Sine wave
- Square wave
- Sawtooth wave
- Pulse wave
- Sound
- Triangle function
- Wave
- Zigzag

References

1. Kraft, Sebastian; Zölzer, Udo (5 September 2017). "LP-BLIT: Bandlimited Impulse Train Synthesis of Lowpass-filtered Waveforms". *Proceedings of the 20th International Conference on Digital Audio Effects (DAFx-17)*. 20th International Conference on Digital Audio Effects (DAFx-17) (<http://www.dafx17.eca.ed.ac.uk/>). Edinburgh. pp. 255–259.



Triangle wave with amplitude = 5, period = 4



Animation of the additive synthesis of a triangle wave with an increasing number of harmonics. See Fourier Analysis for a mathematical description.

- Weisstein, Eric W. "Fourier Series - Triangle Wave" (<https://mathworld.wolfram.com/FourierSeriesTriangleWave.html>). *MathWorld*.
-

Retrieved from "https://en.wikipedia.org/w/index.php?title=Triangle_wave&oldid=1281148272"